κChow-functions

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Abstract.

1. Introduction

1.1. Main results.

2. Framework

2.1. **Posets and Incidence Algebras.** Let $\mathcal{P}$ be a *finite partially ordered set* ranked by the function ρ. We denote by $\mathrm{I}(\mathcal{P})$ the incidence algebra of the poset $\mathcal{P}$ over the ring $\mathbb{Z}[x]$ with the algebra structure given by convolution. Some of the functions we consider in $\mathrm{I}(\mathcal{P})$ are δ; the zeta function, ζ; its inverse, the Möbius function, μ; the characteristic $\mathcal{P}$-kernel χ and its associated functions. Following [FMV24], some of the functions associated to χ are the left and the right Kazhdan–Lusztig–Stanley functions (g and f respectively); the Z function; the reduced characteristic $\mathcal{P}$-kernel $\overline{\chi}$; the characteristic Chow function H; and the left and the right augmented Chow functions (G and F respectively). All these functions live inside a special subalgebra $\mathrm{I}_\rho(\mathcal{P})$ which consists of all the functions $\mathrm{a} \in \mathrm{I}(\mathcal{P})$ that satisfy $\deg(\mathrm{a}_{st}(x)) \leq \rho_{st} := \rho(t) - \rho(s)$ for all $s \leq t$ in $\mathcal{P}$. This subalgebra $\mathrm{I}_\rho(\mathcal{P})$ is equipped with an involution defined by $\mathrm{a} \mapsto \mathrm{a}^{\mathrm{rev}}$ with $\mathrm{a}^{\mathrm{rev}}_{st} := x^{\rho_{st}}\mathrm{a}_{st}(x^{-1})$. We summarize some properties of these functions as follows:

(1) $\chi = \mu \cdot \zeta^{\mathrm{rev}}$ is a $\mathcal{P}$-kernel, i.e. it satisfies $\chi^{-1} = \chi^{\mathrm{rev}}$,
(2) $g = \zeta^{\mathrm{rev}}$; $\zeta^{\mathrm{rev}}_{st} = x^{\rho_{st}}$,
(3) $\mathrm{Z} = g^{\mathrm{rev}} f = g f^{\mathrm{rev}}$ satisfies $\mathrm{Z}^{\mathrm{rev}} = \mathrm{Z}$,
(4) $\overline{\chi}_{st}(x) = \begin{cases} \chi_{st}(x)/(x-1) & \text{if } s < t, \\ -1 & \text{if } s = t, \end{cases}$
(5) $\mathrm{H} := -(\overline{\chi})^{-1}$ satisfies $\mathrm{H}_{st}(x) = x^{\rho_{st}-1}\mathrm{H}_{st}(x^{-1})$,
(6) $F = \mathrm{H} \cdot f^{\mathrm{rev}}$ satisfies $\mathrm{F}^{\mathrm{rev}} = \mathrm{F}$,
(7) $G = g^{\mathrm{rev}} \cdot \mathrm{H}$ satisfies $\mathrm{G}^{\mathrm{rev}} = \mathrm{G}$.

2.2. **The lorem ipsum modules.** Let $R := \mathbb{Z}[x, x^{-1}]$ be the ring of Laurent polynomials with integer coefficients. This ring contains the additive subgroups $\mathrm{Pal}(n)$, for each $n \in \mathbb{Z}$, consisting of all Laurent polynomials p satisfying $p(x) = x^n p(x^{-1})$. We note that these subgroups satisfy $\mathrm{Pal}(n) \cdot \mathrm{Pal}(m) \subset \mathrm{Pal}(n+m)$ and, in particular, $\mathrm{Pal}(0)$ is a ring.

We now consider the module $\mathcal{H}(\mathcal{P})$ of $\mathcal{P}$ which is defined as the free R-module on the set $\{\mathrm{s}\}_{s\in\mathcal{P}}$. A typical element in $\mathcal{H}(\mathcal{P})$ looks like

$$\alpha = \sum_{s\in\mathcal{P}} \alpha_s \cdot \mathrm{s}, \qquad \text{where } \alpha_s \in R.$$

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We also define a family of R-linear map, $\{S_s : \mathcal{H}(\mathcal{P}) \to R\}_{s\in\mathcal{P}}$, given by

$$\sum_{t\in\mathcal{P}} \alpha_t \cdot \mathrm{t} \mapsto \sum_{t\geq s} x^{-\rho_{st}}\alpha_t.$$

Using these maps, we can now introduce some 'local' conditions to impose on the elements $\alpha \in \mathcal{H}(\mathcal{P})$:

(1) $\alpha_s \in \mathbb{Z}[x]$ for every $s \in \mathcal{P}$,
(2) $S_s(\alpha) \in \mathrm{Pal}(0)$ for every $s \in \mathcal{P}$.

Let $\mathcal{H}_p(\mathcal{P})$ the subgroup consisting of all elements $\alpha \in \mathcal{H}(\mathcal{P})$ that satisfy both of the conditions and let $\mathcal{H}'_p(\mathcal{P})$ be the subgroup consisting of only those that satisfy 2. We remark that the subgroup $\mathcal{H}_p(\mathcal{P})$ is a $\mathbb{Z}$-module and the subgroup $\mathcal{H}'_p(\mathcal{P})$ is a $\mathrm{Pal}(0)$-module. To further study these modules, we define two new R-bases of $\mathcal{H}(\mathcal{P})$ given by:

$$\mathrm{kl}^t = \sum_{s\leq t} \mathrm{kl}^t_s(x)\cdot \mathrm{s} := \sum_{s\leq t} x^{\rho_{st}} f_{st}(x^{-2})\cdot \mathrm{s}, \qquad \text{and} \qquad \mathrm{c}^t = \sum_{s\leq t} \mathrm{c}^t_s(x)\cdot \mathrm{s} := \sum_{s\leq t} x^{\rho_{st}} \mathrm{H}_{st}(x^{-2})\cdot \mathrm{s}.$$

We will name the first one as the **Kazhdan–Lusztig basis** and the second as the **Chow basis**. There are a few notes that we can make about these new bases:

(1) $\mathrm{kl}^t \in \mathcal{H}_p(\mathcal{P})$ for all $t \in \mathcal{P}$: We need to show that kl^t satisfies the two conditions. Firstly, since $\deg(f_{st}) < \rho_{st}/2$, we have $\mathrm{kl}^t_s \in \mathbb{Z}[x]$. Secondly, for a fixed $s \in \mathcal{P}$, we have

$$\begin{aligned} S_s(\mathrm{kl}^t) &= \sum_{u\geq s} x^{-\rho_{su}} \mathrm{kl}^t_u(x) \\ &= \sum_{u\geq s} x^{-\rho_{su}} x^{\rho_{ut}} f_{ut}(x^{-2}) \\ &= x^{\rho_{st}} \sum_{u:s\leq u\leq t} x^{-2\rho_{su}} f_{ut}(x^{-2}) \\ &= x^{\rho_{st}} \mathrm{Z}_{st}(x^{-2}). \end{aligned}$$

The factor $x^{\rho_{st}}$ is in $\mathrm{Pal}(2\rho_{st})$ and, since $\mathrm{Z}^{\mathrm{rev}} = \mathrm{Z}$, the factor $\mathrm{Z}_{st}(x^{-2})$ is in $\mathrm{Pal}(-2\rho_{st})$. Thus their product $S_s(\mathrm{kl}^t)$ is in $\mathrm{Pal}(2\rho_{st} - 2\rho_{st}) = \mathrm{Pal}(0)$, as required.

(2) $\mathrm{c}^t \in \mathcal{H}'_p(\mathcal{P})$: This requires $S_s(\mathrm{c}^t)$ to be in $\mathrm{Pal}(0)$ for every $s \in \mathcal{P}$. As in the previous case, $S_s(\mathrm{c}^t)$ now reduces to the product $x^{\rho_{st}} \left(g^{\mathrm{rev}} \cdot \mathcal{H}\right)_{st}(x^{-2}) = x^{\rho_{st}} G_{st}(x^{-2})$ which is also in $\mathrm{Pal}(0)$ since $G^{\mathrm{rev}} = G$.

Since the set $\{\mathrm{kl}^t\}_{t\in\mathcal{P}}$ (resp. $\{\mathrm{c}^t\}_{t\in\mathcal{P}}$) is R-independent, it is also a $\mathbb{Z}$ (resp. $\mathrm{Pal}(0)$)-independent set in $\mathcal{H}_p(\mathcal{P})$ (resp. $\mathcal{H}'_p(\mathcal{P})$). But we can have even more:

Theorem 2.1 *$\{\mathrm{kl}^t\}_{t\in\mathcal{P}}$ forms a $\mathbb{Z}$-basis for $\mathcal{H}_p(\mathcal{P})$ and $\{\mathrm{c}^t\}_{t\in\mathcal{P}}$ forms a $\mathrm{Pal}(0)$-basis for $\mathcal{H}'_p(\mathcal{P})$.*

Proof. We need to show that the sets span the respective modules.

Let $\alpha := \sum_{t\in\mathcal{P}} \alpha_t \cdot \mathrm{t}$ be an element in $\mathcal{H}_p(\mathcal{P}) \subset \mathcal{H}(\mathcal{P})$. In terms of the Kazhdan–Lusztig basis, we write it as $\sum_{t\in\mathcal{P}} \beta_t \cdot \mathrm{kl}^t$. Thus we have

$$\alpha = \sum_{t\in\mathcal{P}} \alpha_t \cdot \mathrm{t} = \sum_{t\in\mathcal{P}} \beta_t \cdot \mathrm{kl}^t = \sum_{t\in\mathcal{P}} \beta_t \cdot \sum_{s\leq t} \mathrm{kl}^t_s \cdot \mathrm{s} = \sum_{s\in\mathcal{P}} \left(\sum_{t\geq s} \beta_t \mathrm{kl}^t_s\right)\cdot \mathrm{s} \implies \alpha_s = \sum_{t\geq s} \beta_t \mathrm{kl}^t_s. \tag{1}$$

We will show that $\beta_s \in \mathbb{Z}$ by induction on the corank of s.

Base case: When $s = \hat{1}$, $\alpha_{\hat{1}} = \beta_{\hat{1}}\mathrm{kl}^{\hat{1}}_{\hat{1}} = \beta_{\hat{1}}$. Since $\alpha \in \mathcal{H}_p(\mathcal{P})$, we have that $S_{\hat{1}}(\alpha) = \sum_{t\geq\hat{1}} x^{-\rho_{\hat{1}t}}\alpha_t = \alpha_{\hat{1}} \in \mathrm{Pal}(0)$. But $\alpha_{\hat{1}}$ also needs to be in $\mathbb{Z}[x]$. Thus, $\beta_{\hat{1}} = \alpha_{\hat{1}} \in \mathrm{Pal}(0) \cap \mathbb{Z}[x] = \mathbb{Z}$ as required.

Inductive step: Assume that for all $t \in \mathcal{P}$ with $\rho(t) > \rho(s)$, we have $\beta_t \in \mathbb{Z}$. Then,

$$S_s(\alpha) = \sum_{t \geq s} x^{-\rho_{st}} \alpha_t.$$

Substituting for α_t using the change of basis formula in 1, we have

$$S_s(\alpha) = \sum_{t \geq s} x^{-\rho_{st}} \sum_{u \geq t} \beta_u \mathrm{kl}^u_t = \sum_{u \geq s} \beta_u \left(\sum_{t : s \leq t \leq u} x^{-\rho_{st}} \mathrm{kl}^u_t \right) = \sum_{u \geq s} \beta_u S_s(\mathrm{kl}^u).$$

Isolating the $u = s$ term, we have,

$$\beta_s = S_s(\alpha) - \sum_{u > s} \beta_u S_s(\mathrm{kl}^u).$$

The entire right-hand-side is in $\mathrm{Pal}(0)$ and thus $\beta_t \in \mathrm{Pal}(0)$.

Using the change of basis formula in 1, we also have $\beta_s = \alpha_s - \sum_{t>s} \beta_t \mathrm{kl}^t_s$. The right-hand-side here is just an expression of polynomials. Combining this fact with the previous, $\beta_s \in \mathrm{Pal}(0) \cap \mathbb{Z}[x] = \mathbb{Z}$.

The proof of the second part is similar. □

We continue our notes about the bases:

(3) $\mathrm{kl}^t_t = \mathrm{c}^t_t = 1 \in \mathbb{Z} \subset \mathrm{Pal}(0)$.

(4) $\mathrm{kl}^t_s \in x\mathbb{Z}[x]$ and $\mathrm{c}^t_s \in \mathrm{Pal}(2)$ for $s < t$ in $\mathcal{P}$: The first part is true because $\deg(f_{st}(x)) < \rho_{st}/2$. For the second assertion, using the fact that $\mathrm{H}_{st}(x)$ is a symmetric polynomial with center of symmetry $(\rho_{st} - 1)/2$, we note that

$$x^{\rho_{st}} \mathrm{H}_{st}(x^{-2}) \in \mathrm{Pal}(2\rho_{st}) \cdot \mathrm{Pal}(-2(\rho_{st} - 1)) \subset \mathrm{Pal}(2).$$

So we have $\mathrm{kl}^t \in \mathbb{Z} \cdot \mathrm{t} + \sum_{s<t} x\mathbb{Z}[x] \cdot \mathrm{s}$ and $\mathrm{c}^t \in \mathrm{Pal}(0) \cdot \mathrm{t} + \sum_{s<t} \mathrm{Pal}(2) \cdot \mathrm{s}$ These decompositions are quite special because the pairs $(\mathbb{Z}, x\mathbb{Z}[x])$ and $(\mathrm{Pal}(0), \mathrm{Pal}(2))$ form complementary subspaces of $\mathbb{Z}[x]$ and $\mathbb{Z}[x, x^{-1}]$ respectively. In other words,

$$\mathbb{Z}[x] = \mathbb{Z} \oplus x\mathbb{Z}[x] \qquad \text{and} \qquad \mathbb{Z}[x, x^{-1}] = \mathrm{Pal}(0) \oplus \mathrm{Pal}(2)$$

as $\mathbb{Z}$-modules and $\mathrm{Pal}(0)$-modules respectively.

References

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